

Colloidal oatmeal: history, chemistry and clinical properties.

Oatmeal has been used for centuries as a soothing agent to relieve itch and irritation associated with various xerotic dermatoses. In 1945, a ready to use colloidal oatmeal, produced by finely grinding the oat and boiling it to extract the colloidal material, became available. Today, colloidal oatmeal is available in various dosage forms from powders for the bath to shampoos, shaving gels, and moisturizing creams. Currently, the use of colloidal oatmeal as a skin protectant is regulated by the U.S. Food and Drug Administration (FDA) according to the Over-The-Counter Final Monograph for Skin Protectant Drug Products issued in June 2003. Its preparation is also standardized by the United States Pharmacopeia. The many clinical properties of colloidal oatmeal derive from its chemical polymorphism. The high concentration in starches and beta-glucan is responsible for the protective and water-holding functions of oat. The presence of different types of phenols confers antioxidant and anti-inflammatory activity. Some of the oat phenols are also strong ultraviolet absorbers. The cleansing activity of oat is mostly due to saponins. Its many functional properties make colloidal oatmeal a cleanser, moisturizer, buffer, as well as a soothing and protective anti-inflammatory agent.